October 28, 2025

Luka Seamus Wright
Oceans Institute
University of Western Australia
Perth WA 6009
Australia
☎ +61 8 6488 8760
✉ luka.wright@research.uwa.edu.au

Dear editor,

I wish to submit a manuscript titled *A model of macroalgal decomposition* for publication in the *Journal of the Royal Statistical Society Series C: Applied Statistics*. I confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

In this paper, I present the first model specifically tailored to macroalgal decomposition, which is notoriously difficult to model. My model builds on the original exponential decay model first described in a highly cited paper on mathematical ecology 62 years ago and the recently developed theory of detrital photosynthesis according to which macroalgal detritus may behave completely contrary to the expectations of the classic model. I illustrate the efficacy of my model on a variety of published empirical datasets using a Bayesian implementation in R with Stan. I anticipate that my model will have far-reaching consequences for the assessment of seaweed blue carbon.

I believe my manuscript is suited to the *Journal of the Royal Statistical Society Series C: Applied Statistics* since it tackles a genuine applied statistical problem in ecology which motivated the work. While I do not develop new statistical theory, my work provides a new mathematical framework to describe the ecological data in question and uses existing advanced statistical methods to build a novel generative model.

All data and code are available in my open-access repository at github.com/lukaseamus/limbodeco and I encourage you to pass these on to the reviewers.

Sincerely,
Luka Seamus Wright